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Temporal patterns of food control outputs and outcomes and effects of targeted control interventions on compliance in food service establishments in Finland

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ABSTRACT

Food control inspections are an essential control measure to ensure compliance, and it is important to understand possible patterns within inspections and their outcomes when developing food control. In this study, we analysed food control inspection reports (n = 86 223) concerning Finnish food service establishments from 2016 to 2021 to reveal temporal patterns in food control outputs and outcomes and the possible reasons behind them, and to evaluate the effects of targeted control interventions on inspection grades. The analyses revealed that the proportions of non-compliances were high throughout the study period in several inspected issues, such as temperature management (7.1–8.1%) and cleanliness (4.4–5.3%). Moreover, the inspection grades improved in only one of sixteen inspection subsections, but in five subsections the proportion of non-compliances increased significantly within the study period (linear regression analysis) and in seven subsections the inspection grades were inferior during the COVID-19 pandemic compared to the pre-pandemic years. In addition to the long-term variation in the level of non-compliances, we observed seasonal variation, with higher proportions of non-compliances in summer compared to other times. This study also showed that targeted food control interventions may influence both the number of inspections and the level of observed non-compliances. We also observed that almost half of the annual pre-planned inspections were made in the proximity of the inspection time in the previous year (44.6% within 30 days), decreasing the surprise factor of the inspections. These results can be used in developing the effectiveness of food control.

1. Introduction

Food business operators (FBOs) are responsible for their procedures and the safety of their food products. EU regulation 852/2004 obligates FBOs to establish and operate food safety programmes and procedures, also called food safety management systems (FSMS), based on hazard analysis and critical control point (HACCP) principles to control potential food safety hazards (EC, 2004). In turn, the role of official food control is to verify FBOs' compliance with food safety regulations (EC, 2004; EU, 2017). Despite food control, non-compliances are common in various food establishment types, and many FBOs fail to correct non-compliances fully or partly within the time limits given by food safety inspectors (Kaskela et al., 2021; Kosola et al., 2022; Läikkö-Roto et al., 2015). Some FBOs have challenges in implementing HACCP principles, such as hazard identification and assessment, for instance due to lack of knowledge and skills, and financial constraints (Baş et al., 2007; Dzwolak, 2019; Herath & Henson, 2010), which may further lead to various non-compliances. Some non-compliances, such as improper food-handling practices and temperature abuse, have been associated with foodborne disease outbreaks (Angelo et al., 2017; EFSA & ECDC, 2022; Moritz et al., 2023; Pihlajasaari et al., 2021). Of all reported strong-evidence foodborne outbreaks in the European Union (EU) in 2021, at least 45% were related to food consumed in restaurants, canteens, and other food service premises, accounting for 57% of all reported foodborne illnesses, 252 hospitalisations, and 7 deaths (EFSA & ECDC, 2022).

To reduce non-compliances and the risk of foodborne outbreaks, the effectiveness of food control should be improved. For that purpose, it is important to monitor indicators of food control, such as inspection results, to detect possible trends and variations from baseline, and then

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react to them effectively when needed (Baert et al., 2011; Codex Alimentarius, 2017; FAO, 2021). EU regulation 625/2017 requires that member states report their food control results annually and consider the inspection results in their regular updates of the multi-annual national control plan. For instance, increased non-compliances on specific areas could lead to targeted inspections related to those issues (DVFA, 2023a). In addition to national reports of inspection results, not many studies have considered compliance development over time. Long-term studies may reveal patterns not detected in annual monitoring. Evaluating food control results over time gives important information, but to further understand the behaviour of FBOs in certain activities, Ridderstaat and Okumus (2019) suggested that the seasonality of inspection results could also be considered in forthcoming studies. Both long-term and seasonal studies demand standardised inspection system so that the inspection results are comparable with each other.

In Finland, the food control inspection system is harmonised, which provides good conditions for long-term monitoring. The Finnish food control system follows the requirements of EU and national food safety regulations (EU, 2017; Food Act, 2021), and food control, except for control of slaughterhouses, is conducted by 60 local food control units (FFA, 2023c) that are supervised by six regional units and a central authority, the Finnish Food Authority (FFA). Along with many other countries, the food control results are disclosed in Finland (Aik et al., 2018; DVFA, 2023b; FFA, 2024; Fleetwood, 2019). The disclosure system, called Oiva, was established in May 2013 in the food service sector and is coordinated by the Finnish Food Authority, which provides detailed evaluation guidelines for inspectors (FFA, 2023a). The guidelines include inspection subsections (e.g., actions and training of personnel) consisting of a variable number of specific items (e.g., hand hygiene) to be inspected. All inspected items are graded using a four-level scale: A = excellent, complies with requirements; B = good, minor issues that do not impair food safety or mislead consumers; C = to be corrected, impairing food safety or misleading consumers; and D = poor, jeopardising food safety or considerably misleading consumers, or failing to correct non-compliances. The grades of inspection subsections, as well as the overall grade, are determined by the lowest grade of the individual items. One or more items receiving grade C or D leads to a follow-up inspection in which at least the correction of observed non-compliances is evaluated. All relevant inspection items should be inspected at least once every three years (FFA, 2023a). The recommended inspection frequency in food service sector varies from one inspection in three years to three inspections per year, depending on the type of operation and the number of meals served per month (FFA, 2023b). All inspection data is collected in the same information management system throughout the country.

Along with the regular pre-planned food control inspections, the Finnish Food Authority coordinates priorities and projects focusing on certain items (FFA, 2022a). Similar control interventions, called campaign inspections, are conducted in, for instance, Denmark (DVFA, 2023a). The targeted food control interventions may focus on one or more issues with more in-depth inspection and guidance (DVFA, 2023a; FFA, 2022a). The Finnish Food Authority typically offers education related to the targeted issues for inspectors (FFA, 2022a). Educational interventions for inspectors have been demonstrated to improve food safety inspectors' knowledge and performance in implementing legislation on food safety (Arnold et al., 2020). In addition to increasing inspectors' and FBOs' knowledge of the targeted inspection areas, increasing the number of inspections related to rarely-inspected items can also be an aim of the targeted interventions (FFA, 2021). It is important to analyse the effect of targeted control interventions on compliance to develop food control in the future.

Food control should contain an element of surprise to create a truthful picture of the inspected establishments. Inspections should therefore be performed unannounced when possible (EU, 2017). Studies have observed inspection grades to be lower, particularly in easy-to-correct inspection items, in unannounced inspections compared to pre-announced inspections in restaurants and other food establishments (Kaskela et al., 2021; Kosola et al., 2022; Nwako, 2017; Waters et al., 2013a). Furthermore, Makofske (2019, 2021) found that even when the inspections are unannounced, the level of compliance is affected by FBOs' ability to anticipate the inspection timing when multiple restaurants and retail food markets in the same building are inspected on the same day. To better understand FBOs' ability to anticipate inspection timing, the timings of regular inspections should be studied.

The aims of this study were to 1) assess the compliance level of food service establishments over time and seasonally to recognise temporal patterns and the potential reasons for them; 2) investigate the effects of targeted food control interventions on inspection results and number of inspections; and 3) study the proximity of annual inspections between years. We hypothesised that 1) temporal patterns in compliance can be observed; 2) the targeted food control interventions affect the inspection results and the number of inspections; and 3) inspections are clustered at the same time of year as in the previous year. Revealing unrecognised patterns in compliance, knowledge of effects of interventions, and the timing of annual inspections is useful for authorities preparing future risk-based inspection plans and targeted interventions.

2. Materials and methods

2.1. Food control inspection report data

The materials for this study included food control inspection reports from all inspected food service establishments in Finland between 2016 and 2021. A total of 86 223 pre-planned (91%) and follow-up (9%) inspections were carried out at 39 838 food service establishments, representing five establishment types (Table 1). Inspection reports from the first years (2013–2015) of the Finnish disclosure system were not included to exclude possible discrepancies in the inspection grading during the run-in phase of the system. The inspection reports were acquired from the Finnish Food Authority.

2.2. Food control interventions

Information on targeted food control interventions, including the time and aims, target food business sector(s), and participating control units, was gathered from the Finnish Food Authority's websites (FFA, 2022b) and reports (FFA, 2019a; 2019b, 2021; 2022a). Interventions were included in the study if they met the following inclusion criteria: 1) the intervention occurred between 2016 and 2021; 2) the target food sectors included food service; 3) the intervention focused on specific inspection item(s); and 4) the participating local control units were specified. Eventually, three control interventions concerning four inspection subsections and six inspection items altogether were included. All the included control interventions were also targeted at other food sectors in addition to food service, but in this study, we analysed their effects on detected compliance and number of inspections specifically in the food service sector. The first intervention occurred in 2018 (from March to October) and focused on foodstuffs with protected status (FFA 2019a, b). A total of 14 local control units took part in the intervention, but two of them did not carry out inspections at food service establishments and were therefore excluded from our analyses. The second intervention was a two-year-long (2020–2021) control priority targeted at marketing (FFA, 2021; 2022a). The third intervention was also a two-year-long (2020–2021) control priority that targeted maintenance (items "condition of premises" and "working utensils, fixtures and equipment") and cleanliness (items "general order and cleanliness of facilities" and "cleanliness of working utensils, equipment and cleaning equipment") (FFA, 2021; 2022a). Data concerning maintenance and cleanliness were analysed separately, but the inspection items within them were combined in the analyses. All local control units took part in the second and third control interventions. The aims of the interventions

Table 1
Descriptive statistics of food control inspections of food service establishments in Finland in 2016–2021.

Type of establishment	Number of inspections (%)	Mean number of inspections per year (min–max)	Grades[a] (%)			
			A	B	C	D
Restaurant	36 327 (42.1)	6055 (4984–6733)	32.9	46.9	19.4	0.8
Institutional catering[b]	32 582 (37.8)	5430 (3975–6354)	58.4	35.3	6.2	0.1
Cafe	10 164 (11.8)	1694 (1175–2118)	46.1	42.2	11.2	0.5
Grill or fast-food	6189 (7.2)	1032 (830–1225)	41.2	42.3	15.7	0.7
Pub	961 (1.1)	160 (59–321)	50.4	41.7	7.4	0.5
Total (all establishments)	86 223 (100)	14 371 (11 056–16 475)	44.9	41.5	13.0	0.5

[a] A = Excellent, B = Good, C = To be corrected, D = Poor.
[b] Includes industrial kitchens (e.g., schools, hospitals), central kitchens, kitchens that prepare precooked food products for service, and catering services.

included reducing misleading of consumers (first and second intervention) and promoting food safety (third intervention) as well as harmonisation of control. Education concerning each intervention was provided to inspectors.

2.3. Statistical analyses

Statistical analyses were conducted using SPSS 29 (IBM SPSS Statistics, USA) unless otherwise stated. Statistical significance was accepted with a p-value <0.05. Grades C (To be corrected) and D (Poor) were referred to as non-compliances.

2.3.1. Temporal analyses of compliance and inspections

We analysed annual numbers of inspections and inspected establishments, and the annual overall grade distributions of inspections and inspected establishments considering only the lowest grade received in each year. Furthermore, we used linear regression to analyse the development of the proportion of C and D grades within inspections over the years for overall grades and for inspection subsections. In addition, we analysed the significance of the differences in grades (A and B compared to C and D) between the pre-COVID-19-pandemic (2016–2019) and pandemic (2020–2021) years with Pearson's Chi-squared test or Fisher's exact test when any of the expected cell frequencies was less than five. We considered only the lowest grade of inspected items per subsection within each inspection in the analyses. One subsection, marketing requirements, was excluded from the analyses, as it had not been inspected every year. Statistical analyses of overall grades were conducted for all inspections and separately for different establishment types.

The number of inspections was also explored by month to observe possible monthly patterns. To investigate possible monthly patterns in non-compliances, we used seasonal index analysis, which eliminates long-term trends as well as cyclic and irregular components from the data and reveals the seasonal component. The seasonal index indicates how much higher (index >1.0) or lower (index <1.0) a value of interest usually is at a particular time (e.g., a month) compared with other times (e.g., all months of the year) (Siegel & Wagner, 2023). We calculated monthly seasonal indices for the proportion of overall grades C and D (henceforth "CD-proportion") using data from January 2016 to December 2021. The indices were calculated for all food service inspections, and separately for all five establishment types. In addition, the seasonal indices were calculated by excluding inspections made in establishments that had been inspected only in the summer (June–August, $n = 8700$) to evaluate the effect of establishments that operate only in the summer on the level of non-compliance. To calculate the monthly seasonal indices, the CD-proportion was calculated for each month in each year. Then, a centred moving average of 12 months was calculated for each month (moving average for the first and last six months of the raw data are unavailable). Then, a ratio-to-moving-average was calculated by dividing the original CD-proportion of the month by the centred moving average. Finally, the seasonal index for each month was calculated by averaging ratio-to-moving-averages of the month between different years (Siegel & Wagner, 2023). The indices were calculated using Microsoft Excel (2023).

In addition to monthly investigation of non-compliances, we compared CD-proportions between summer months (June–August) and other times of the year for overall inspection grades and for all inspection items that had been inspected at least a thousand times in 2016–2021. To evaluate the significance of the differences detected, we used Pearson's Chi-square test or Fisher's exact test when any of the expected cell frequencies was less than five. The analyses were also performed for establishment types with all items included.

2.3.2. Intervention analysis

Pearson's Chi-square test was used to determine if there were statistically significant differences in the inspection results (grades A and B compared to grades C and D) before, during, and after (if available) the intervention, over three consecutive years. Bonferroni correction was used in pairwise comparisons.

2.3.3. Interval between pre-planned inspections

To assess the proximity of annual inspections between years, we analysed time intervals between pre-planned inspections in establishments that had been inspected once per year in 2016–2019 ($n = 1485$). Years 2020 and 2021 were excluded to avoid the effect of the COVID-19 pandemic on the results. The time intervals were calculated between inspection dates in each two consecutive years (2017–2016, 2018–2017, and 2019–2018) for each establishment, resulting in a total of 4455 intervals.

3. Results

3.1. Development of compliance and number of inspections over time

The annual CD-proportions of overall grades did not show a significant linear change during the study time, but the grades were significantly inferior during the pandemic compared to the pre-pandemic years ($p < 0.001$) both for all inspections and for inspected establishments with only the poorest annual grades included in the analyses. Moreover, the overall grade distributions varied between establishment types, restaurants having the highest CD-proportions (Table 1, Supplementary Fig. S1). When establishment types were examined separately, no significant linear changes were observed in the annual CD-proportions, and significantly inferior grades during the pandemic years compared to the pre-pandemic years were observed only in restaurants ($p = 0.004$).

Furthermore, the linear regression analysis revealed that the CD-proportion decreased significantly in one and increased in five of the sixteen inspection subsections during the study period (Table 2). The inspection grades of the five subsections were also significantly inferior during the pandemic years compared to the pre-pandemic years. In addition, the subsections "actions and training of personnel" and "food testing" received inferior grades during the pandemic ($p < 0.001$ and $p = 0.016$, respectively). The highest annual CD-proportions were seen in

Table 2
Proportion of inspection grades C and D in food service establishments in Finland from 2016 to 2021.

Inspection subsection[a]	2016	2017	2018	2019	2020	2021	*p*-value[b]	Mean number (min–max) of inspections per year
Overall grade[c]	13.6	13.1	12.5	14.0	13.2	15.3	0.24	14 371 (11 056–16 475)
Food temperature management[d]	8.1	7.6	7.3	8.1	7.1	8.0	0.78	13 755 (10 432–15 977)
Own-check	6.0	5.7	5.1	4.9	4.1	4.3	0.002	9964 (7388–12 028)
Suitability, adequacy, and maintenance of facilities and equipment	5.7	5.0	4.6	5.6	4.5	5.2	0.53	11 748 (9503–13 497)
Actions and training of personnel	4.8	3.5	3.4	4.3	4.7	5.3	0.32	13 468 (10 101–15 901)
Sale and serving[d]	4.5	4.7	4.1	4.9	4.4	4.9	0.53	10 769 (7656–13 021)
Cleanliness of facilities, surfaces, and equipment	4.5	4.4	4.5	5.3	4.4	5.0	0.35	13 809 (10 463–16 043)
Food testing	2.9	2.4	2.1	2.6	2.6	3.4	0.39	3766 (2946–4218)
Deliveries of food	2.6	2.5	2.3	2.3	2.1	2.9	0.91	7337 (5525–9030)
Information provided on foods	1.1	1.7	1.8	2.3	2.6	4.4	0.01	5601 (4435–6880)
Hygiene of food production or handling	1.0	2.0	2.0	3.9	2.3	2.5	0.23	3179 (2362–3979)
Display of the inspection report	1.0	1.0	1.0	1.0	1.2	1.4	0.04	9370 (6731–11 003)
Special requirements for specific food products	0.7	0.4	2.0	7.5	7.9	10.8	0.003	309 (137–399)
Packaging materials and food contact materials	0.5	0.7	0.7	0.8	0.9	1.1	0.001	4474 (3368–5372)
Substances causing allergies or intolerances	0.4	0.4	0.5	0.6	0.5	0.4	0.62	7566 (5407–8989)
Traceability and recalls	0.4	1.0	1.0	1.2	1.5	1.2	0.04	4195 (3198–5786)
Composition of foodstuffs	0.0	0.0	0.0	4.3	0.0	9.1	0.12	30 (11–55)

[a] Only subsections inspected every year are included.
[b] Linear regression.
[c] The overall grade is determined by the lowest grade of all inspected subsections at the inspection. Proportion of grade D varies from 0.4 to 0.6%.
[d] "Food temperature management" relates to storage temperatures, whereas management of service temperatures were evaluated in subsection "sale and serving".

the subsection "food temperature management" (7.1–8.1%) in most years (Table 2).

Between 2016 and 2021, the total number of inspections decreased from 16 475 to 12 136 and the proportion of inspected establishments from 41% to 32%. The biggest decreases coincided with the implementation of a new data management system in 2019 and the pandemic in 2020 (Fig. 1).

3.2. Seasonal patterns of non-compliances and inspections

The seasonal index analyses revealed a seasonal pattern in the CD-proportion of overall grades. The analyses showed that the CD-proportion was higher in summer (June–August, index 1.13–1.17) and lower in October–May (index 0.92–0.96) (Fig. 2). Furthermore, excluding establishments inspected only in the summer (representing food establishments operating during summer months) increased the indices for all summer months (Fig. 2) and all establishment types (Supplementary Fig. S2).

Of all inspection items with a minimum of 1000 inspections in 2016–2021 (n = 37), 23 items' CD-proportions differed significantly (p

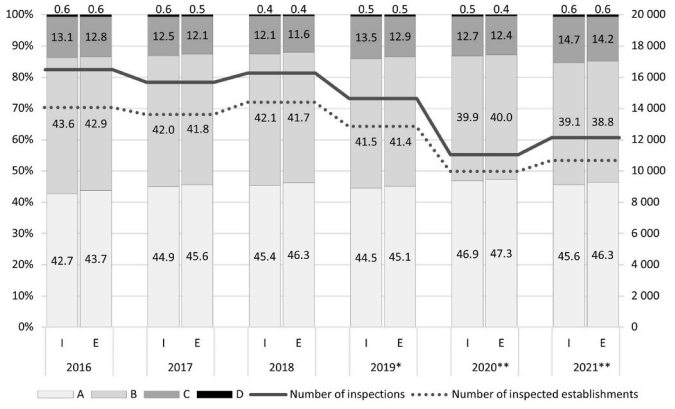


Fig. 1. Overall grade distribution (%) of inspections (I) and inspected food service establishments (E), and the number of inspections and inspected food service establishments in 2016–2021. The overall grade of establishment represents the lowest grade received during the year. *Introduction of new food control data management system (VATI). **Years of COVID-19 pandemic.

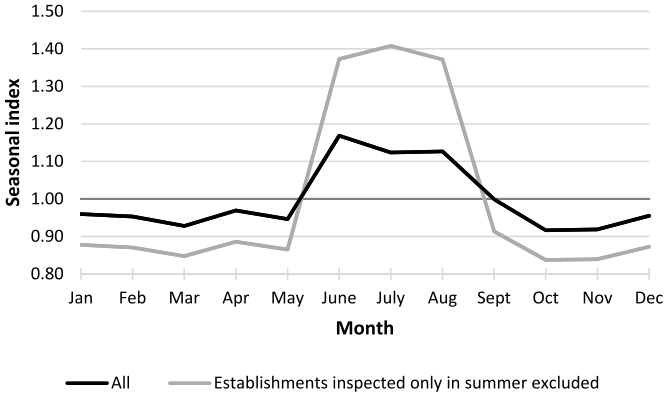


Fig. 2. Seasonal index of the proportion of overall grades C and D in all food service establishments and with exclusion of establishments inspected only in summer (June–August) in 2016–2021. A higher index indicates a higher proportion of C and D grades.

< 0.05) between summer and other times of the year (Table 3). In these 23 items, the ratios of CD-proportions (summer/other time) varied from 1.15 to 1.87. When the same analyses were done for all five establishment types separately, significant differences were found in fifteen items in one to four establishment types, most commonly in institutional catering (Supplementary Table S1).

In 2016–2019 the number of inspections was 17–30% lower in January, July, and December compared to the average of all months. In 2020, the first year of COVID-19 pandemic, the number of inspections decreased drastically, particularly in March–May (Fig. 3).

3.3. Effects of targeted food control interventions

Different changes in the inspection grades and numbers of inspections were detected during and/or after the interventions (Fig. 4.). The CD-proportion increased significantly the year after the intervention concerning foodstuffs with protected status (from 0.8% to 9.6%, p = 0.001). However, the CD-proportion concerning maintenance and cleanliness decreased significantly during the first year of the intervention (from 2.1% to 1.7% in maintenance, p = 0.021; and from 3.1% to 2.4% in cleanliness, p < 0.001), but returned to the previous level

Table 3
Overall grade and inspection items receiving significantly lower grades during summer (June–August) compared to other times of the year in food service establishments in Finland in 2016–2021.

Inspection item	Ratio of the proportion of C and D grades (summer/other time)	*p*-value[a]	Number of inspections[b]
Vermin and other animals	1.87 (1.1/0.6)	<0.001	35 990
Temperature management of quick-frozen and frozen foodstuffs	1.70 (4.3/2.5)	<0.001	6770
Marketing	1.66 (3.1/1.9)	0.04	3034
Display of the inspection report	1.65 (1.6/0.9)	<0.001	56 219
Monitoring of employees' health status	1.54 (7.6/4.9)	<0.001	27 767
Waste management	1.48 (0.7/0.5)	0.01	39 199
Verification of hygiene proficiency	1.47 (1.9/1.3)	<0.001	45 636
Management of shelf-life and sale period of products in sale of foods	1.44 (3.9/2.7)	<0.001	12 261
Work clothes	1.39 (1.4/1.0)	<0.001	68 510
Cold-stored foodstuffs	1.38 (6.9/5.0)	<0.001	79 443
Hot-stored foodstuffs	1.35 (3.9/2.9)	<0.001	21 713
Suitability of premises for use as food premises	1.33 (3.9/3.0)	<0.001	61 308
Management of shelf-life and sale period of products in serving of foods	1.33 (5.7/4.3)	<0.001	56 827
Mandatory food information	1.32 (2.9/2.2)	<0.001	32 325
Reception of foods	1.30 (2.8/2.2)	<0.001	41 305
Sampling and own-check testing[c]	1.27 (3.1/2.5)	0.01	29 726
General order and cleanliness of facilities	1.26 (3.9/3.1)	<0.001	81 034
General compliance of own-check[d]	1.25 (4.0/3.2)	<0.001	92 430
Hand hygiene	1.20 (1.6/1.3)	0.01	74 790
Chilling	1.20 (10.5/8.8)	<0.001	31 576
Cleanliness of working utensils, equipment, and cleaning equipment[e]	1.19 (2.5/2.1)	<0.001	145 826
Storage of foodstuffs	1.18 (1.8/1.5)	0.01	69 781
Suitability and maintenance of working utensils, fixtures and equipment	1.15 (2.1/1.8)	0.03	62 994
Overall grade	1.20 (15.6/12.9)	<0.001	86 233

[a] Pearson's Chi-square test.
[b] Items with more than 1000 inspections included.
[c] Item "actions taken based on test results" was combined into this item in 2020. The two items are combined in this table.
[d] Two other items were combined into this item in 2020. All three items are combined in this table.
[e] Item "cleanliness of the space for cleaning equipment and cleaning equipment" was combined into this item in July 2021. The two items are combined in this table.

during the second year. No significant changes in the CD-proportion were detected for marketing. However, the number of inspections of marketing multiplied drastically during the intervention (Fig. 4b). Furthermore, the number of inspections concerning foodstuffs with protected status also increased (Fig. 4a) but inspections concerning maintenance and cleanliness decreased during the intervention (Fig. 4c and d).

3.4. Intervals between pre-planned inspections

Intervals between annual pre-planned inspections in 2016–2019 varied between 57 and 690 days (mean = 377 and median = 371 days). Out of all 4455 intervals, 14.2% varied by seven days or fewer, and 44.6% varied by thirty days or fewer from one year (Supplementary Fig. S3). When examining the interval means per establishment, they varied between 262 and 479 days (median 374 days). In 57 (3.8%) establishments, one (n = 56) or two (n = 1) of the three intervals were exactly 365 days.

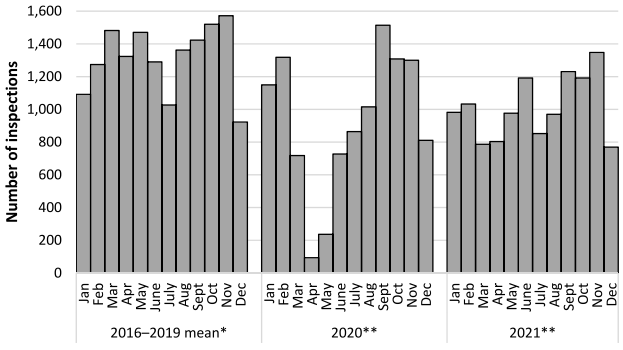


Fig. 3. Monthly number of inspections of food service establishments before (*) and during (**) COVID-19 pandemic.

4. Discussion

Our data showed that the food safety level measured with the overall inspection grade was quite stable over 2016–2021 as 12.5–15.3% of inspections revealed non-compliances. The Finnish Food Authority has stated that the level of food safety in Finland is good (FFA, 2024). However, this does not mean that food safety could not be improved. It is of concern that over one-tenth of food service establishments have non-compliances that decrease or endanger food safety annually, and that some establishments fail to take corrective actions after identified non-compliances. The reasons behind non-compliances were not studied here, but other studies suggest that different factors, such as poor attitude towards food hygiene, lack of time, and weak financial situation, may cause non-compliances and possibly prevent taking corrective actions in food businesses (Kettunen et al., 2018; Kiviniemi et al., 2024; Yapp & Fairman, 2006). Our analyses revealed that the proportions of non-compliances were somewhat high throughout the study period in several inspected issues, such as temperature management (7.1–8.1%) and cleanliness (4.4–5.3%). Non-compliances in both of these have been associated with foodborne disease outbreaks (Leinonen et al., 2022; Petran et al., 2012). Therefore, it would be important to reduce these non-compliances.

There are studies suggesting better food safety levels, measured by inspection outcomes, in some other countries compared to Finland. For instance, in Denmark (Vainio et al., 2023), Austria (Lueckl et al., 2019), and Toronto, Canada (Young et al., 2023) the proportion of inspections with identified non-compliances have been lower. However, comparing results between different countries is very challenging, as their inspection systems and evaluations differ from each other. Still, it would be important to be able to perform comparative studies between countries with similar inspections systems as this could lead to observations that would improve the effectiveness and consistency of food control. To make international comparisons of food control outcomes easier and to further develop food control, it would be important to assess whether there are elements in food control and reporting of the outcomes that

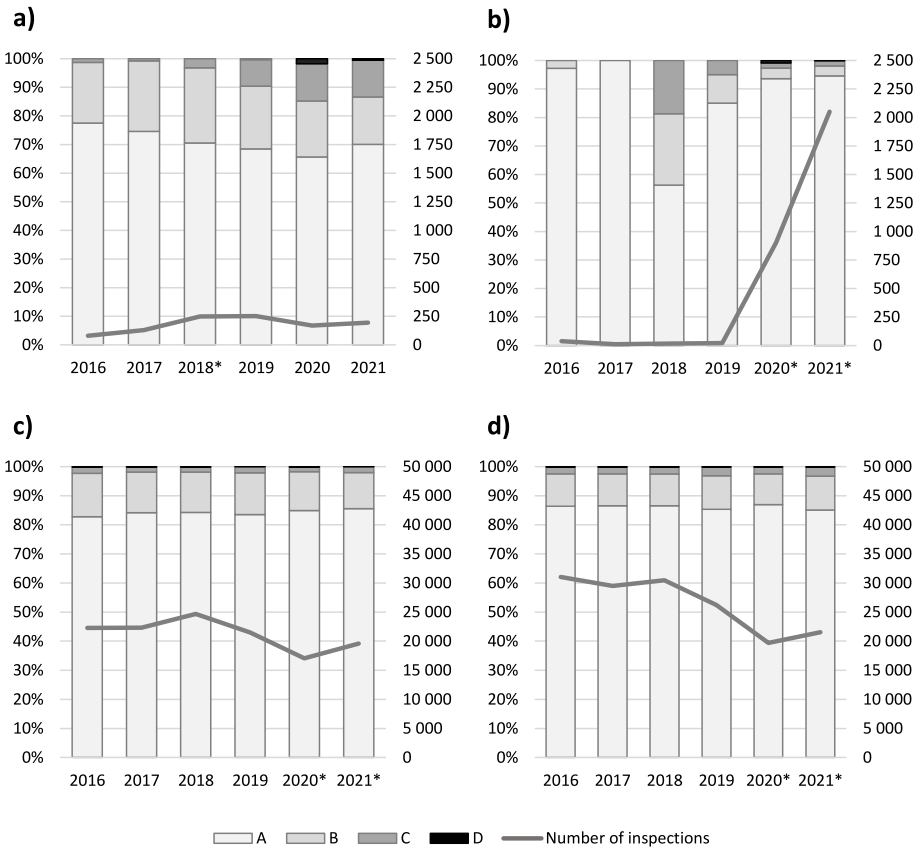


Fig. 4. Grade distributions and numbers of inspections of items included in control interventions. a) Foodstuffs with protected status, 12 local units, b) Marketing, all local units. c) Maintenance (of facilities and equipment), all local units, d) Cleanliness (of facilities, surfaces and equipment), all local units. *Year(s) of the control intervention.

could be harmonised at the EU level.

Although the overall inspection grades remained somewhat stable, the analysis of the inspection subsection grades showed that almost one-third of the subsections had linearly deteriorating grades during the study period. Many of these subsections were information-related, such as "information provided on foods", "display of the inspection report", and "special requirements for specific food products". The deterioration of grades regarding the last is most probably related to the intervention concerning foodstuffs with protected status, as the majority of the inspections within the subsection were related to that item. However, the deteriorating grades of other subsections cannot be explained with national targeted interventions. This may indicate that Finnish food control has increased emphasis on information-related subjects, resulting in more identified non-compliances, or that food businesses' compliance has genuinely decreased. It is of concern if consumers are not provided correct or sufficient information on foods as it may cause food safety risks for allergic people (Gallo et al., 2020) or economic loss if consumers are misled into paying higher prices for foods.

In addition to linear changes, we observed that many subsections, such as "actions and training of personnel", "packaging materials and food contact materials", and "food testing", had inferior inspection grades during the COVID-19 pandemic compared to the pre-pandemic years. The differences concerned restaurants in particular. Possible reasons for increased non-compliances during the pandemic include labour shortages and financial issues, which worsened in the Finnish food service sector during the pandemic (Karikallio & Arovuori, 2023). Labour shortages may cause working in haste and the employment of workers with poor food safety knowledge, which may both increase non-compliances in several issues (Kouabenan & Ngueutsa, 2016; Yapp & Fairman, 2006). In addition, financial problems caused by the pandemic may have led to neglect of requirements to save costs.

FBOs have been obligated to establish, maintain, and provide evidence of an appropriate food safety culture (FSC) since 2021 (EU, 2021). An appropriate FSC is important because it enhances food safety by increasing the awareness and improving behaviour of employees (EU, 2021). It will be interesting to see whether this change in legislation will have an impact on FSC and therefore compliance in food businesses.

This study highlights the decrease in inspection numbers, which may have consequences on food control and food safety. The decrease in inspections was 26% during the study period, but the decrease in inspection numbers has already been ongoing for much longer, as in 2009 there were over 40 000 inspections in the food sector (Evira, 2010), and now the number is half of that (FFA, 2024). There has been little research on how the number of inspections influence compliance in food establishments, although inspections are a fundamental tool for food control. However, the decreased number of inspections has led to increased inspection intervals, which have been observed to be associated with inferior inspection grades in food production establishments in Finland (Kosola et al., 2022). Therefore, it is possible that decreasing numbers of inspections decrease the level of food safety, although more research is needed to understand the connection between the number of inspections and compliance among FBOs.

The decrease in inspection numbers is a result of many factors. For instance, the recommended inspection frequencies were decreased in Finland in 2016, and a new management system for food control data was implemented nationally at the beginning of 2019, but it faced critical technical problems soon after its implementation (Kapanen, 2019). While digital systems may be beneficial for food control in many ways (Grau-Noguer et al., 2023), the implementation and management of new systems should not take resources away from the basic tasks of food control. Furthermore, the pandemic decreased the number of inspections, particularly in 2020. Thus, the increase of non-compliances,

especially in 2021, could to some extent be related to the decreased inspection numbers during the pandemic. However, the effects of the pandemic on food establishments' compliance may be multifaceted, as Young et al. (2023) also observed a decrease in the inspection numbers during the pandemic, but instead of a decrease in compliance, they reported that inspection outcomes were better compared to the pre-pandemic level for 2 years, after which they started to worsen again. Furthermore, as compared to some other countries, such as Denmark, where the inspection frequency is reduced when FBOs voluntarily implement a certified food safety management system (FSMS) (DVFA, 2023a), the status of food establishments' FSMSs does not automatically affect inspection numbers in Finland (Piira et al., 2021).

In addition to annual changes in inspection outcomes, this study revealed a seasonal pattern in non-compliances, as they were most common during the summer. This phenomenon was observed in almost all types of food service establishments, but especially in institutional catering. Moreover, our analyses showed that establishments that operate only in summer do not explain the increase of non-compliances in summer, thus the seasonal pattern appears to be a genuine phenomenon in year-round operating food service establishments. Findings by Wong et al. (2015) suggest similar seasonality in overall grades in restaurants in New York City, and Young et al. (2023) also found inspection pass rates in restaurants and take-out establishments to be lowest in August in Toronto. Other studies comparing compliance in temperature control between summer and other seasons have (Baldera Zubeldia et al., 2016; Waters et al., 2013b) or have not (Waters et al., 2013a) detected increased non-compliances in summer. To broaden the understanding of seasonality in compliance, we analysed seasonal differences in multiple items related to a wide range of topics. As we detected an increase in non-compliances in summer in various areas, there seem to be several reasons explaining the seasonality in compliance. For example, non-compliances related to temperature control (Dominianni et al., 2018) and vermin control (Walker et al., 2003) could be caused by higher outdoor temperatures influencing the operations in the premises. Non-compliances concerning food temperature control cause increased food safety risks, as temperature control is important regarding the safety of foods (Angelo et al., 2017; EFSA & ECDC, 2022; Petran et al., 2012).

Furthermore, we detected increased non-compliances in summer in over a dozen inspection items that are likely not related to the outdoor temperature, such as hand hygiene and storage of foodstuffs, which are also important for the prevention of foodborne illnesses (Angelo et al., 2017; Kosola et al., 2023). One possible cause for the increase of non-compliances in the wide range of topics in the summer could be related to the use of short-term employees in food service. Employees working for a period of less than three months are not obligated to pass the official hygiene passport test, which has been shown to be important in increasing the food safety knowledge of young persons (Vaarala et al., 2021). The situation could possibly be improved by requiring employees to have a hygiene passport even at the start of their employment, although this could cause challenges in recruiting short-term employees. Another possible reason for more non-compliances in summer could be increased customer volume and rush (Kouabenan & Ngueutsa, 2016) in summer compared to other seasons.

The present study shows that targeted interventions may affect not only the number of inspections, but more importantly also compliance. The biggest change in compliance was seen in the intervention concerning protected status of foodstuffs, which was a rarely inspected topic. The identified non-compliances increased during the intervention and for two years after, after which they decreased somewhat. It seems that the inspectors' ability to identify non-compliances increased during the intervention. Furthermore, it is possible that FBOs increased compliance in response to the higher identification level, as was reported in a study from the USA (Jin & Lee, 2014). Consequently, as the changes in the sensitivity of identification of non-compliances were more evident in rarely inspected items, it is important to continue with such control interventions. However, according to a survey conducted by Pham et al. (2012), some inspectors may lack confidence in their knowledge of commonly inspected items, such as cleanliness of utensils and equipment. Thus, education on these items should be offered to inspectors as well.

This study shows that a remarkable part of annual inspections take place at almost the same time of the year as the previous inspection, which can have consequences on the outcome of the inspection. If the timing of inspections in an establishment varies little, inspectors are not able to detect possible seasonal changes in the establishment, and the unexpectedness of inspections may become impaired. The unexpectedness of inspections is important in acquiring a truthful picture of the establishments (Makofske, 2019, 2021) and for detecting food fraud (Joenperä & Lundén, 2023). Therefore, the timing should be considered, particularly regarding establishments that are inspected only once a year or less frequently. Furthermore, this study shows that the numbers of inspections were lowest in January, July, and December, meaning that the probability of getting inspected then is smaller. These findings were not surprising, because inspectors typically take time to make food control plans in January, and July and December are central holiday months in Finland. However, the fact that non-compliances are more common during summertime and that the number of inspections then is fewer together create a concerning situation, which should be acknowledged among food control authorities and FBOs, for example through targeted inspections and education of food service workers.

The inspection data used in this study was extensive, covering all food control inspections concerning food service establishments in Finland over a period of six years. However, this study has some limitations. First, the analyses did not consider some background variables that can influence the compliance of food establishments (e.g., the existence of a certified FSMS in the inspected establishments), and therefore the results do not enable interpretation of the influence of such factors on compliance. Secondly, all but one of the studied interventions occurred during the COVID-19 pandemic, which decreased inspections and possibly influenced compliance, making the interpretation of intervention results more complicated. The study period did not include the time after these interventions, thus we were not able to analyse their possible effects on the inspection grades and numbers in the long term. Furthermore, the exclusion method used in the seasonal index analyses (to see the possible effects on compliance of establishments that operate only in the summer) may have caused some erroneous exclusion of establishments that operate year-round but have been inspected only during summertime. However, this does not cause systematic bias in the results, as the unintentionally excluded year-round operating establishments are most likely similar on average to the establishments that remained in the analyses.

5. Conclusions

This study showed that there are both long term and seasonal patterns in the compliance level of food service establishments in Finland, as non-compliances have increased in several inspection subsections linearly, and they have been most common in summer in various issues. While the COVID-19 pandemic seems to have affected compliance negatively, no specific explanation for the linear changes before the pandemic was identified. Furthermore, we detected that non-compliance levels among some issues highly relevant to food safety, such as temperature management, have been consistently high advocating for the need of improved compliance. Further research is needed to assess whether and how the compliance level and food safety could be improved.

Moreover, we found that targeted food control interventions may increase, decrease, or have no effect on non-compliances in food service sector. However, the results suggest that targeted control interventions are beneficial, especially regarding rarely inspected issues, as they appear to affect the sensitivity to identify non-compliances in those

issues. Finally, this study revealed that almost half of annual inspections are temporally clustered between years, which may reduce the surprise factor of the inspections and prevent inspectors from detecting possible seasonal changes in the establishments. Therefore, it would be important to avoid carrying out annual inspections repeatedly at the same time of year. The results of this study are useful for authorities preparing future inspection plans and targeted interventions.

CRediT authorship contribution statement

Paula Saarimaa: Writing – original draft, Investigation, Formal analysis, Conceptualization. **Mikko Kosola:** Writing – review & editing, Data curation. **Mari Nevas:** Writing – review & editing, Supervision. **Janne Lundén:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodcont.2024.110710.

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